

~~chain rule~~
~~pf~~

when.

$$\begin{array}{l} y = f(x) \\ u = g(x) \end{array} \quad \text{where}$$

$$\begin{array}{l} y = f(x) \begin{cases} \Delta u \\ \Delta y \end{cases} \\ u = g(x) \begin{cases} \Delta x \\ \Delta u \end{cases} \end{array}$$

$$\Rightarrow \frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \quad (\Delta u \neq 0)$$

But.

$$\lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} = \frac{dy}{du}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} = \frac{du}{dx}$$

$$\Delta x \rightarrow 0 \Rightarrow \Delta u \rightarrow 0$$

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \right)$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x}$$

$$= \lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x}$$

$$= \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= f'(u) \cdot g'(x)$$

$$= f'(g(x)) \cdot g'(x)$$